

- 3** A student measures the current and the potential difference for a resistor in a circuit.

$$\text{current} = (50.00 \pm 0.01) \text{ mA}$$

$$\text{potential difference} = (500.0 \pm 0.1) \text{ mV}$$

The measurements are used to calculate the resistance of the resistor.

What is the percentage uncertainty in the calculated resistance?

- A** 0.0002%      **B** 0.0004%      **C** 0.02%      **D** 0.04%